RAMYA K

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PROFESSIONAL SUMMARY:

- Full-Stack Software Developer with 5+ years of experience in designing and deploying enterprise applications using Java, Spring Framework, and Microservices architectures. Skilled in backend and frontend development, focusing on building scalable, high-performance solutions.
- Proficient in Microservices design and development, utilizing Spring Boot for service implementation, Spring Cloud Gateway for API routing and load balancing, and Eureka for service discovery.
- Highly skilled in working with relational and NoSQL databases, including MySQL, MongoDB, Oracle and PostgreSQL. Experienced in writing complex SQL and PL/SQL queries, optimizing stored procedures, and ensuring high data integrity and performance in Oracle-backed systems.
- Experienced in ORM technologies such as Hibernate and Spring Data JPA, with a focus on transaction management, and optimized querying.
- Expert in designing and implementing RESTful and SOAP-based APIs using Servlets and JAX-WS, ensuring seamless data exchange via JSON and XML.
- Proficient in implementing logging, monitoring, and distributed tracing with tools such as Zipkin to enhance observability, troubleshooting, and overall system performance in microservices-based applications.
- Designed JWT, OAuth2 for secure authentication and Resilience4J for circuit breaking and fault tolerance.
- Experienced in front-end development with React and Angular (v8+ with RXJS), building dynamic, component-based UIs and integrating them with backend microservices for a seamless user experience.
- Experienced in integrating LLM-based AI solutions into microservice architectures using LangChain, OpenAI APIs, and vector databases like Pinecone and FAISS for intelligent semantic search and contextual query resolution.
- Proficient in DevOps practices, with expertise in GitHub Actions, Docker, Kubernetes, and AWS services (EC2, ECR, RDS) for cloud-based deployment, management, and scaling.
- Proficient in writing unit tests using JUnit, along with integration testing and mocking using Mockito, ensuring code reliability and quality.
- Experienced in automated UI testing using Selenium WebDriver, and service virtualization with CA DevTest for end-to-end application testing and faster QA cycles.
- Experienced in Agile environments, collaborating with cross-functional teams to deliver high-quality software through iterative development and continuous feedback.

TECHNICAL SKILLS:

Programming Languages	Java (8+), C, C++, Python, JavaScript (ES6+), TypeScript.
Java Technologies	JSP, Java Servlets, Java Reflection API, JDBC, JPA, JTA, JMS, JAX-WS, Project Reactor
Backend & Microservices	Spring (Core, Boot, Data JPA, Security, AOP), Spring Boot, Hibernate, REST APIs, GraphQL, Node.js, Fast API, Spring WebFlux, Resilience4J.
Databases	PostgreSQL, MySQL, MongoDB, Oracle (PL/SQL), DynamoDB, Cassandra
Frontend	React.js, Angular 14 (RxJS, Reactive Forms, Angular CLI, Angular Material), HTML, CSS, PHP, JavaScript, TypeScript, Node.js.
Tools and Software	Git, Maven, Gradle, IntelliJ, Visual Studio, Visual Studio Code, Spring Tool Suite (STS), Android Studio, Jira, Confluence, Agile/Scrum, Tomcat, Web Sphere.
DevOps	Jenkins, Git, GitHub Actions, Azure DevOps, Docker, Kubernetes, Terraform.
Monitoring & Quality	SonarQube, Nexus Repository, Zipkin, Splunk.
Cloud Platforms	AWS (EC2, VPC, RDS, S3, Lambda, CloudWatch), Azure (Functions, Event Hubs, Databricks, DevOps, Pipelines, Blob Storage, Key Vault, Application Insights, Azure Monitor).
Testing Tools	JUnit, Mockito, SOAPUI, Selenium WebDriver, Postman, Jest, Playwright, Cypress, JMeter, Service Virtualization.
Messaging Systems	RabbitMQ, Kafka.
AI & LLM Technologies	LangChain,Retrieval-Augmented Generation (RAG), OpenAI APIs, HuggingFace Transformers, Pinecone, FAISS, Azure ML, TensorFlow, Keras, Vector Search.

PROFESSIONAL EXPERIENCE:

Capital One, **Dec 2023 - Present**
Java Full Stack Developer
Responsibilities:

- Designed and developed cloud-native microservices for digital banking and payments, supporting 400K+ active retail users, focusing on secure transaction processing, API integration, and high system availability. Worked on backend services that supported customer features such as fund transfers, payment scheduling, and account activity monitoring. Ensured reliability and scalability by adopting event-driven architecture, implementing strong security measures, and integrating services seamlessly with both frontend applications and downstream systems.
- Developed microservices using Spring Boot, focusing on key areas such as transaction management with Spring Data JPA and Hibernate for database interactions.
- Designed and implemented Spring Cloud Gateway for API routing and Eureka for service discovery, facilitating efficient microservices communication.
- Developed and optimized GraphQL queries alongside REST APIs, cutting average response times by ~25%.
- Integrated circuit breaking using Resilience4J for fault tolerance, ensuring seamless communication during failures and improving system reliability.
- Leveraged Java 11 features such as var, enhanced String methods, and the standardized HTTP Client API to improve code maintainability and performance.
- Built asynchronous services using Spring WebFlux, improving throughput and reducing thread usage by ~30% under peak load.
- Implemented Kafka for asynchronous communication between services, ensuring efficient message processing.
- Integrated JWT and OAuth2 for secure authentication and role-based authorization, ensuring secure API access.
- Contributed to front-end modules in React, Redux, and Angular (v8+ with RXJS), ensuring responsive and dynamic user interfaces.
- Implemented state management in React applications using Redux, employed Redux Thunk for asynchronous data fetching, and optimized component re-rendering for a smooth user experience.
- Managed containerization using Docker and orchestrated with Kubernetes for deploying services in cloud environments.
- Integrated Zipkin for distributed tracing and Splunk for centralized logging and real-time monitoring, improving observability, issue diagnosis, and performance troubleshooting across microservices.
- Investigated and resolved production issues reported by internal users and automated alerts, often involving service timeouts, failed Kafka consumers, or misconfigured API gateways.
- Monitored reactive service flows and performance with Spring Boot Actuator and AWS CloudWatch and used integrated logs with Splunk dashboards to trace issues in asynchronous microservices, coordinating quick fixes and redeployments via GitHub Actions and Docker.
- Handled on-call responsibilities during support hours, prioritizing incident tickets and working with QA/DevOps teams to ensure service SLAs were met without affecting customer-facing operations.
- Deployed applications on AWS EC2, used RDS for database management, and ECR for container image storage, ensuring scalable and high-performance cloud infrastructure.
- Collaborated on CI/CD automation with GitHub Actions and Azure DevOps, supporting cloud-native deployments and reducing deployment time by ~40% and minimizing release rollbacks.
- Designed and optimized PostgreSQL and Aurora schemas, queries, and indexes to support transaction-heavy workloads. Improved query execution times by reducing full table scans, applying partitioning strategies, and tuning indexes for high-volume payment data.
- Leveraged MongoDB and Cassandra to support event-driven microservices, enabling distributed storage of real-time transaction and customer activity data. Implemented sharding and replication strategies to ensure high availability and low-latency access across regions.
- Utilized DynamoDB for serverless, key-value storage to back low-latency services such as session management and caching layers. Configured auto-scaling and optimized read/write capacity to handle spikes in usage during peak transaction hours.
- Integrated reporting workflows into Java-based services using REST triggers and automated pipelines, enabling real-time operational dashboards and compliance-driven reporting for business stakeholders.
- Developed and deployed Retrieval-Augmented Generation (RAG) pipelines using LangChain, LlamaIndex, OpenAI APIs, and HuggingFace Transformers to integrate internal data sources, enabling domain-specific semantic search and contextual response generation across enterprise workflows.
- Built integration modules bridging Python-based LLM workflows with existing Spring Boot microservices, enabling end-to-end support for natural language queries via REST APIs.
- Extended backend systems to consume vector search results and generate context-aware outputs, supporting both internal operations and customer-facing services.
- Tuned chunking strategies, prompt structures, and retrieval parameters for RAG pipelines, reducing LLM response latency by ~20% and improving accuracy by ~15% based on feedback and observability metrics.

Environment: Java 8/11, Spring, Spring Boot, Spring WebFlux, PostgreSQL, MongoDB, Cassandra, RESTful APIs, GraphQL, JavaScript, TypeScript, Angular(v8+), React, Redux, Apache Kafka, Docker, Kubernetes, GitHub Actions, Jenkins, JWT, AWS, Resilience4J, LangChain, LlamaIndex, OpenAI APIs, FAISS, Pinecone, Python.

Tata Consultancy Services, **Apr 2021 – Jul 2022**
Software Developer
Responsibilities:

- Modernized large-scale telecom subscription and billing platform supporting 600K+ accounts, transforming legacy monolithic applications into a microservices-based architecture. Refactored core modules, built new service components, and implemented seamless integration between legacy SOAP systems and modern RESTful APIs to improve scalability, maintainability, and system reliability.
- Developed the application at the service layer using Java features like Streams, Functional Interfaces, and Lambda expressions, along with core Java concepts such as Exception Handling, Multithreading, and Collections to improve performance and reliability of subscription and billing workflows.
- Implemented data access and business logic layers with Spring and Hibernate, using HQL, Criteria queries, and batch processing for large datasets. Optimized performance through second-level caching to reduce the number of hits to configuration table data and transaction management, reducing database load and improving response times by ~30%.
- Implemented Dependency Injection (DI) with annotations and integrated Spring AOP for cross-cutting concerns like logging and error handling, enhancing maintainability and code quality.
- Developed RESTful APIs with Spring Boot for subscription and billing modules, while maintaining SOAP-based integrations (JAX-WS) for interoperability with legacy telecom systems.
- Built React.js dashboards that surfaced real-time subscription and billing insights with clean API integrations, while also contributing Angular (v8) components to existing modules to maintain feature parity and consistent UX. Reduced dashboard load times by ~20%, improving usability for internal agents.
- Applied JUnit for unit testing and Mockito for mocking dependencies, ensuring high test coverage and application reliability.
- Deployed and supported applications in Tomcat, using Log4j for monitoring, troubleshooting, and performance analysis in production-like environments.
- Added structured logging with Log4j (MDC/correlation IDs) and lightweight health/metrics endpoints for subscription/billing services, enabling trace-through across REST and SOAP calls and reducing incident triage time by ~25% during invoice runs and plan-change bursts.
- Developed and optimized MySQL/PostgreSQL SQL (indexes, execution-plan review, batch queries) to support invoice generation, proration, and regulatory/aging reports, stabilizing long-running jobs and reducing contention by ~35% during nightly billing windows.
- Participated in Agile development practices, including sprint planning, daily standups, and code reviews, to ensure timely and high-quality delivery. Managed source code with Git, ensuring version control and efficient collaboration across teams.
- Participated in daily triage calls to discuss ongoing production issues, coordinated with cross-functional teams for quick patch deployment, and maintained clear documentation of root causes and resolution steps.
- Automated ETL pipelines using Python and FastAPI, processing millions of records daily and reducing manual data handling time by 60%. This freed up data engineers to focus on more strategic tasks, improving overall productivity and efficiency of the team.
- Implemented sandboxed NLP prototypes (FastAPI + transformer models via OpenAI/Hugging Face) enabling semantic search over knowledge bases and auto-tagging of support tickets, with benchmarks on precision/recall and latency to guide next-step pilots.

Environment: Java 8, Spring, Spring AOP, Spring Data JPA, MySQL, PostgreSQL, JDBC, Hibernate, REST API, SOAP, Junit, Log4j, Mockito, Tomcat, Postman, Html, CSS, JavaScript, React.js, Angular(v8), Python, FastAPI, OpenAI API, Hugging Face Transformers, Git.

Vitech Systems Group **Jan 2020 – Mar 2021**
Junior Software Developer
Responsibilities:

- Worked on a policy administration platform used by 50K+ policyholders, enabling customers to create accounts with policy numbers, manage multiple policies per account, and generate certificates, building core services with Spring Boot and integrating dashboards with React.
- Gathered system requirements and collaborated with the business team to refine the requirements and the Software Requirement Specification (SRS) document, as well as the Architecture document.
- Contributed to Object-Oriented Analysis and Design (OOAD) using UML, creating Class Diagrams, Sequence Diagrams, and State Diagrams to enhance system architecture.

- Developed backend services using Spring Boot, focusing on RESTful API implementation and seamless integration with MySQL databases, improving API reliability and reducing error rates by ~15%.
- Performed database tasks including queries, schema updates, and integrity checks to support production workflows.
- Participated in debugging sessions and log analysis (Log4j) to identify and resolve issues, improving system reliability and performance.
- Developed unit tests using JUnit and performed API testing with Postman to validate application functionality, increasing test coverage to ~70% and reducing regression defects.
- Gained hands-on experience with version control in Git, managing branches and commits to collaborate effectively within the team.
- Built responsive and user-friendly frontend components using React, ensuring a seamless user experience by integrating backend services with RESTful APIs. Optimized component rendering and state management, reducing dashboard load times by ~25%.
- Documented task progress and shared key learnings with the team, fostering collaboration and knowledge sharing.

Environment: Java 8, Spring Boot, Maven, HTML, CSS, React.js, REST API, MySQL, Hibernate, Jira, JUnit, Mockito, Postman, Git, Log4j.

EDUCATION:

- Master of Science in Computer Science, State University of New York at New Paltz - May 2024